PAPI: Tackling NVIDIA and AMD Emerging Architectures

Treece Burgess

Scalable Tools Workshop
July 26-30, 2026

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THE UNIVERSITY OF TENNESSEE KNOXVILLE

PAPI

- PAPI offers a **universal interface** and **methodology** for monitoring **performance counters** from diverse **hardware and software components**

WHY DOES IT MATTER?

- **If you can’t measure it, you can’t improve it!**
- PAPI enables app developers to see, in near real time, the relation between **SW performance** and **HW events** across the entire compute system

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- **AMD CPUs** up to Zen 5
- **ARM, ARM64** up to ARMv8, ARMv9 (including Cortex and Neoverse series)
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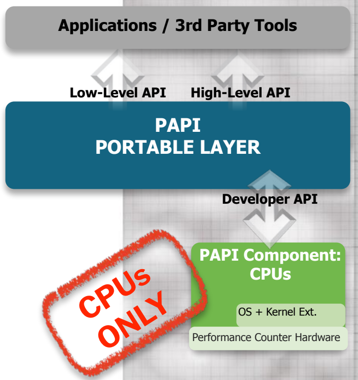
1999 - 2009



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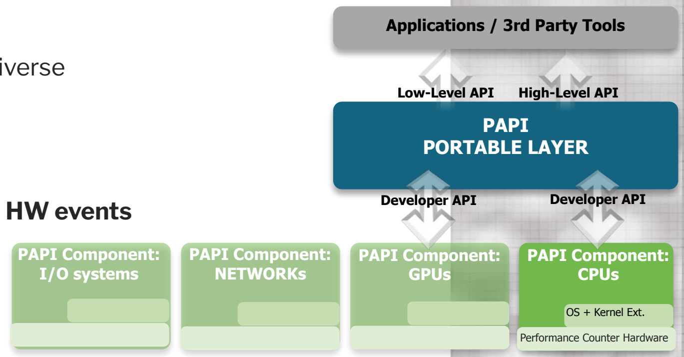
2009 - 2018



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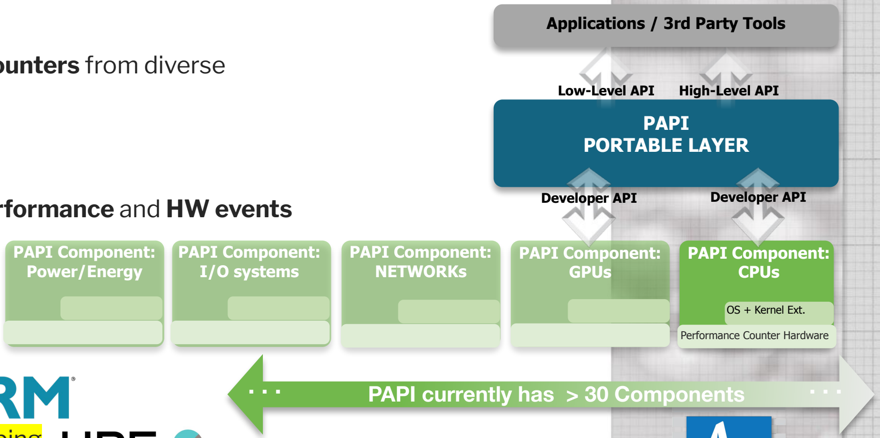

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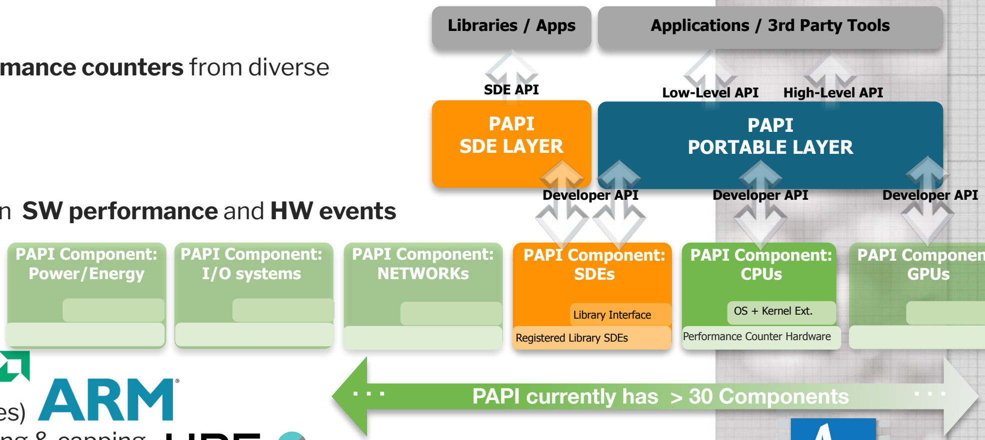

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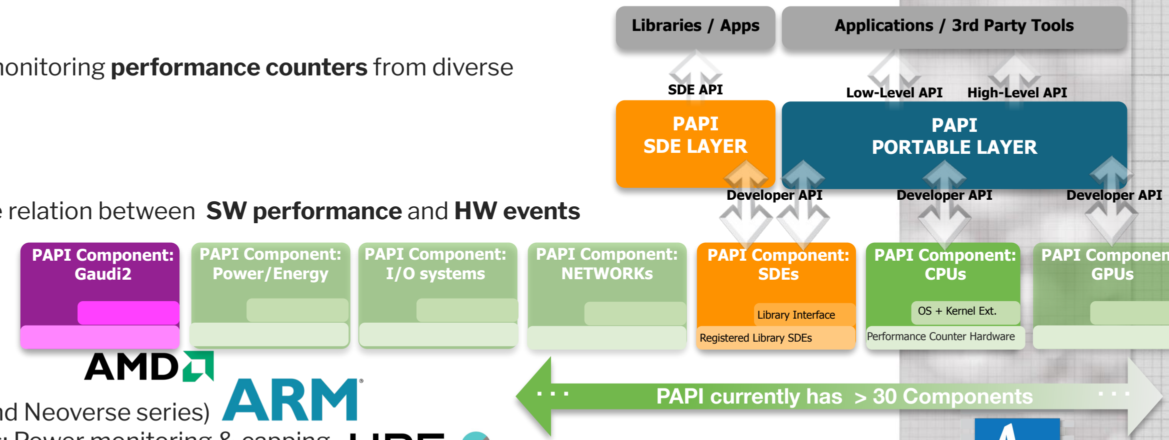

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Tackling NVIDIA's Emerging Architectures

Juggling CUPTI APIs

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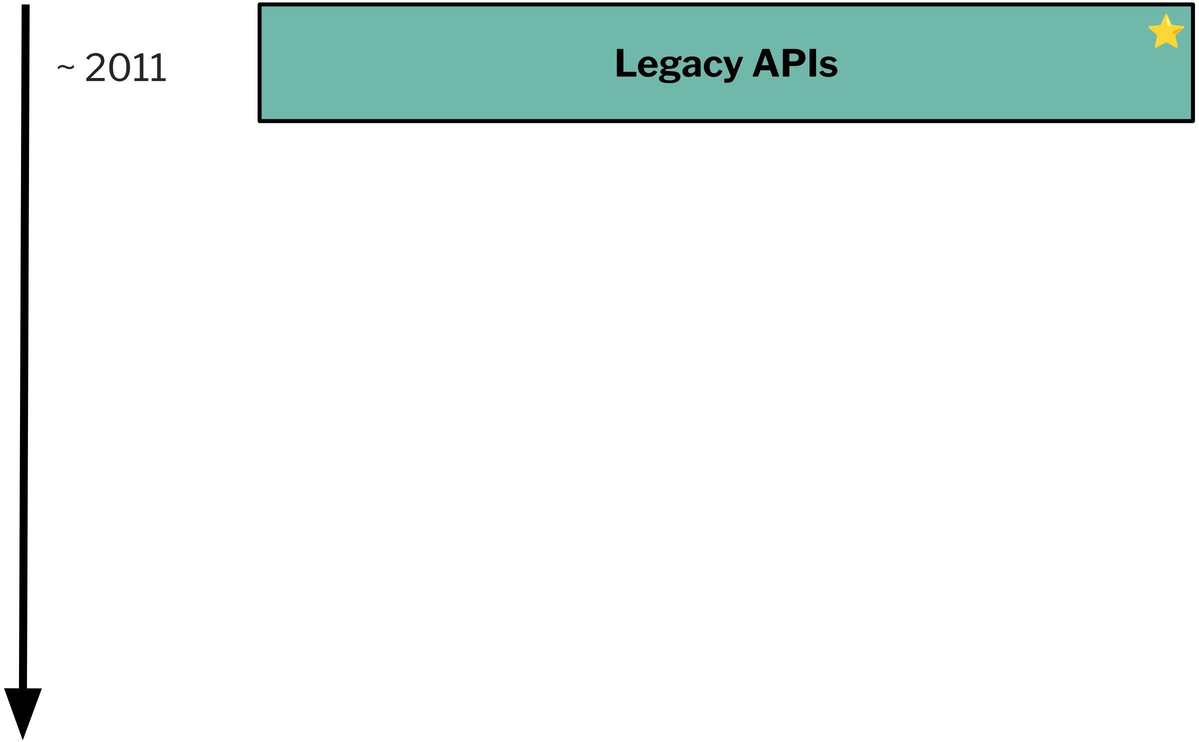
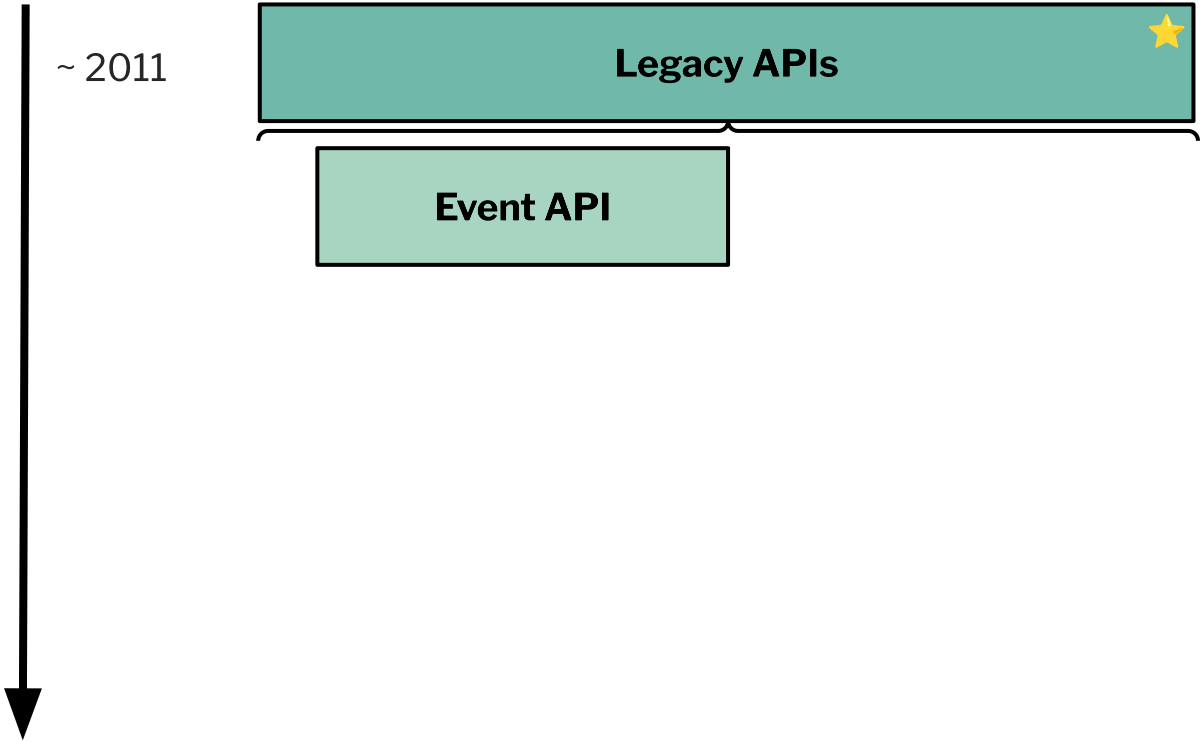


Figure 1: Timeline of CUPTI APIs (⭐ - Supported in PAPI)

Juggling CUPTI APIs



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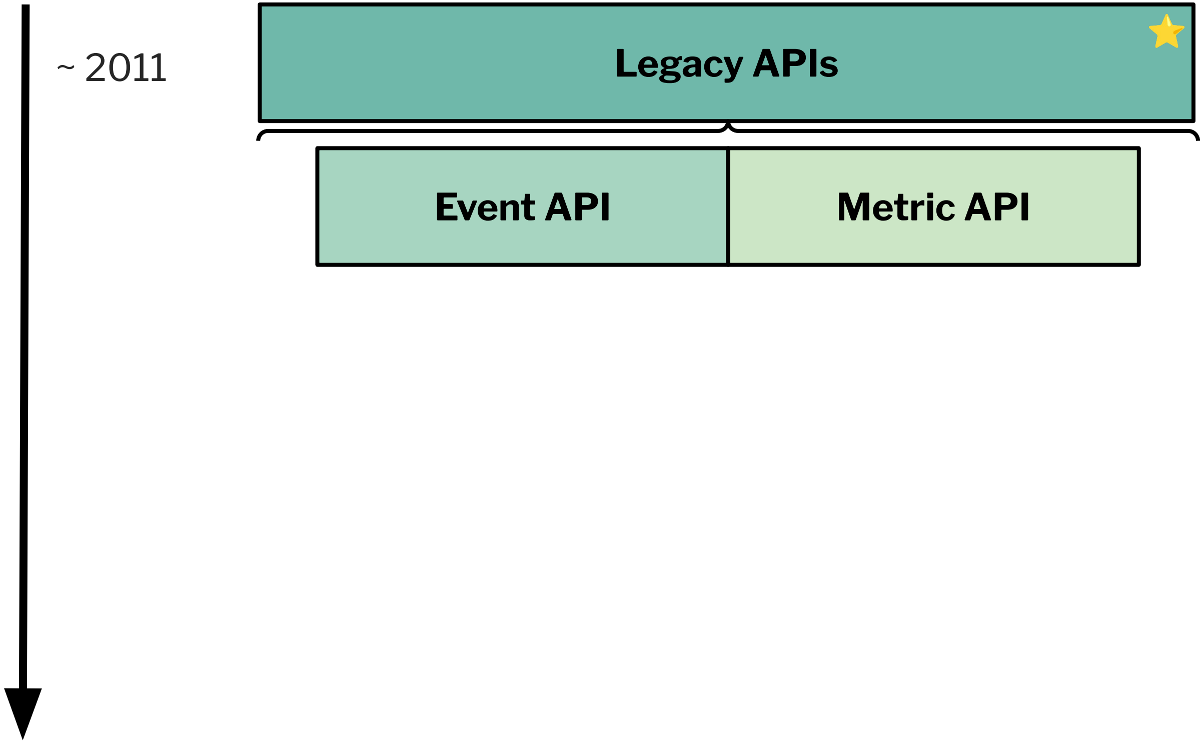

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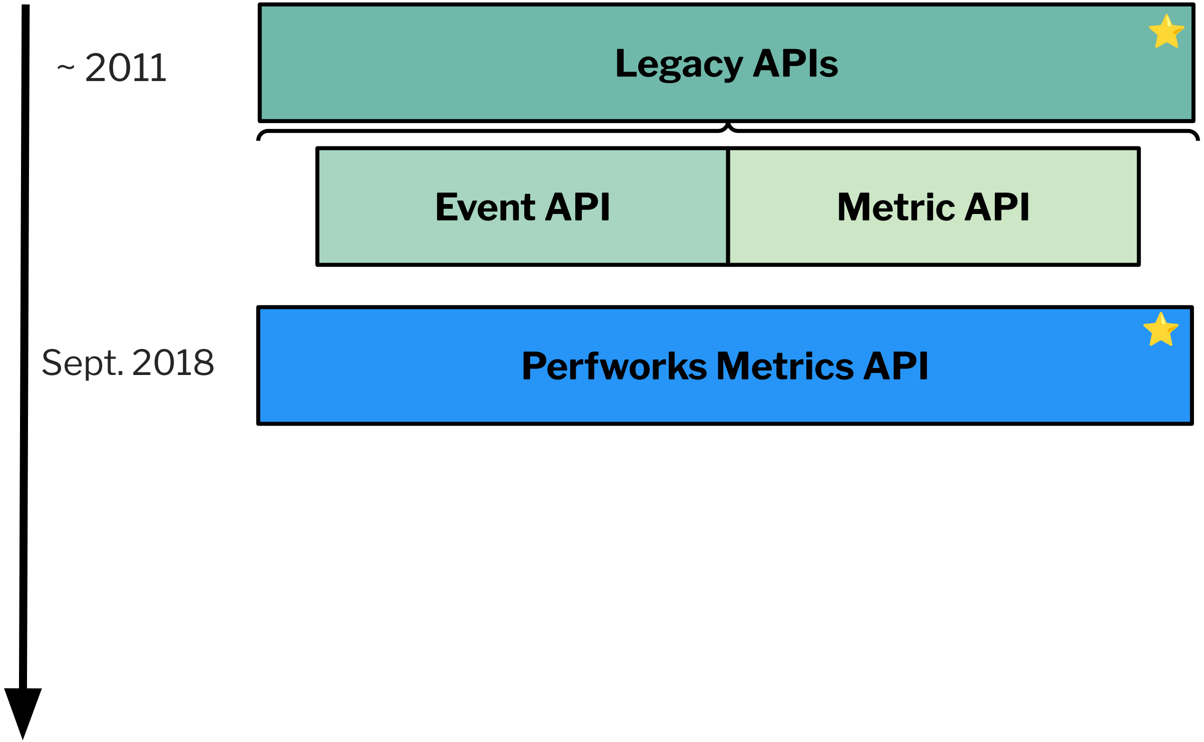

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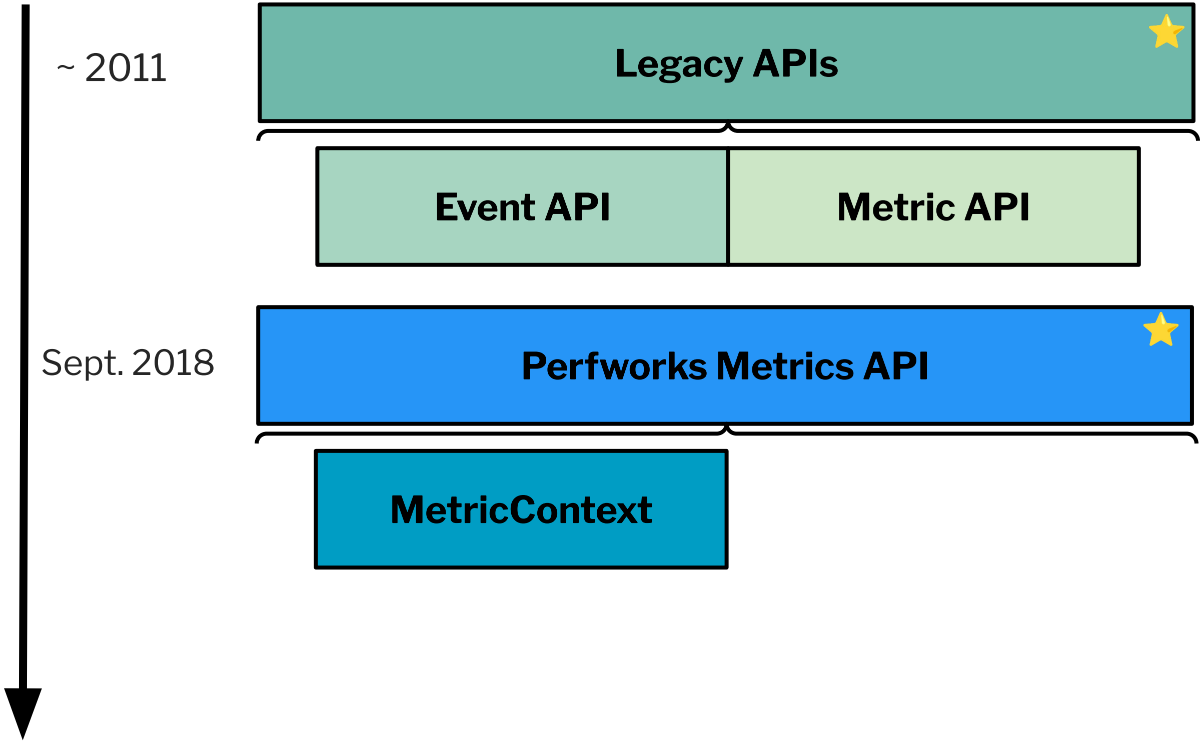

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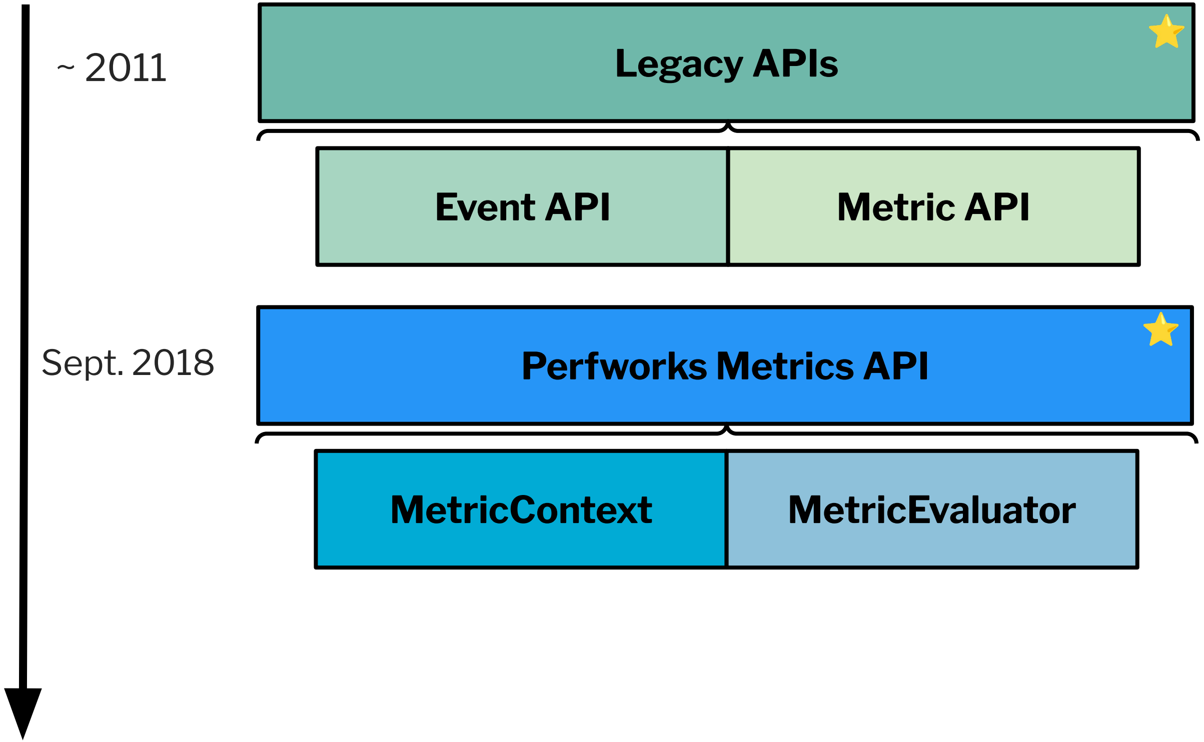

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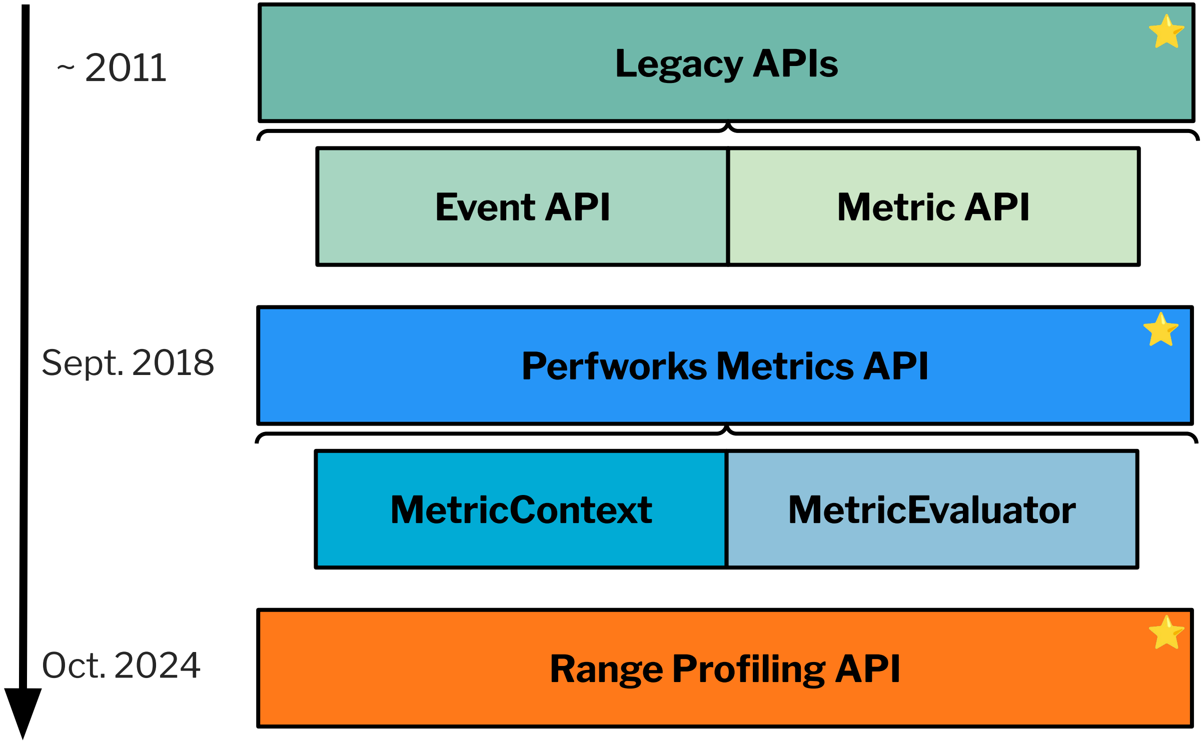

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Mapping CUPTI APIs to PAPI Components

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cuda component

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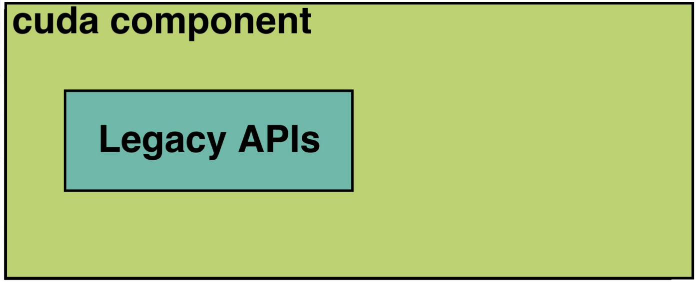

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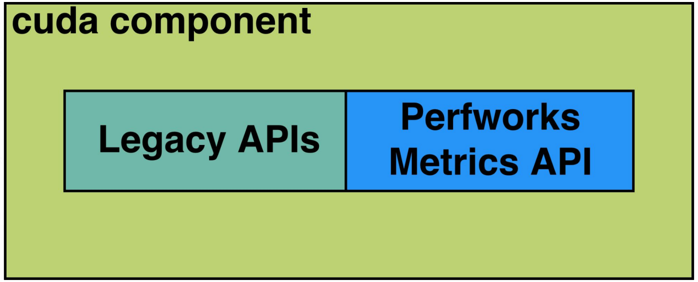

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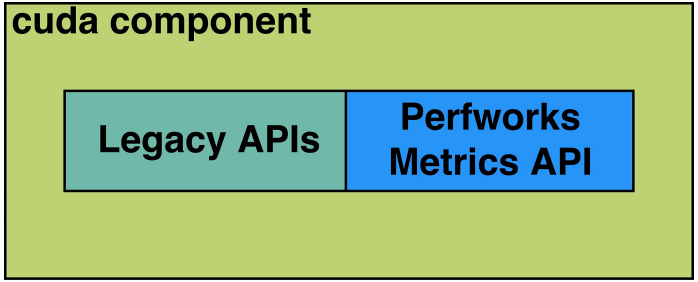

Notable additions:

- Device (***:device=***) and statistic (***:stat=***) qualifiers
  - Reduced the native event count from hundred of thousands to tens of thousands (e.g. 300K to 60K on an NVIDIA H100)
  - Cuda component handles default cases:

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//User application code.
PAPI_add_named_event(..., "cuda:::dram__bytes");
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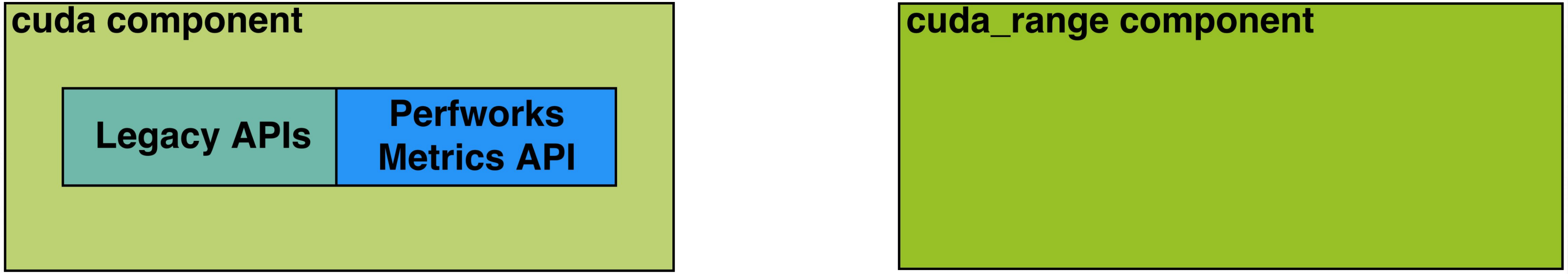

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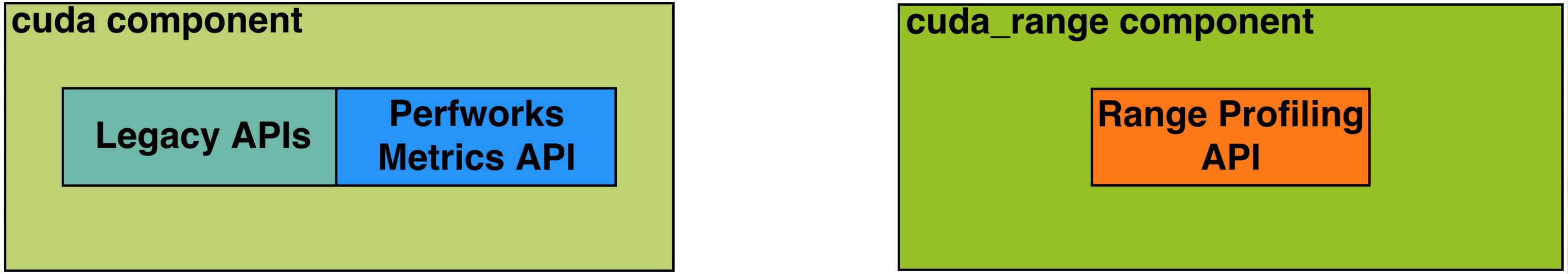

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- **Legacy APIs**
- **Perfworks Metrics API**

cuda_range component

- **Range Profiling API**

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Notable additions:

- Submetric (***:submetric=***) qualifier
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Cuda Toolkit Support

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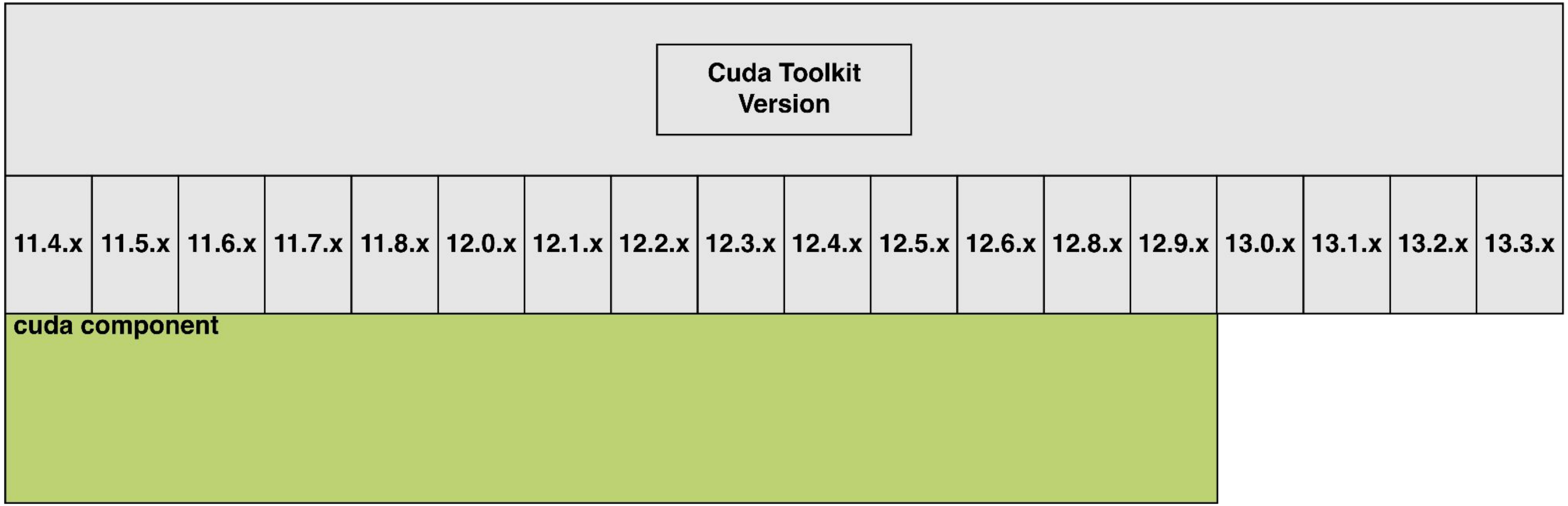
Cuda Toolkit Version																	
11.4.x	11.5.x	11.6.x	11.7.x	11.8.x	12.0.x	12.1.x	12.2.x	12.3.x	12.4.x	12.5.x	12.6.x	12.8.x	12.9.x	13.0.x	13.1.x	13.2.x	13.3.x

Figure 2: Component Breakdown of Supported Cuda Toolkit Versions

Cuda Toolkit Support



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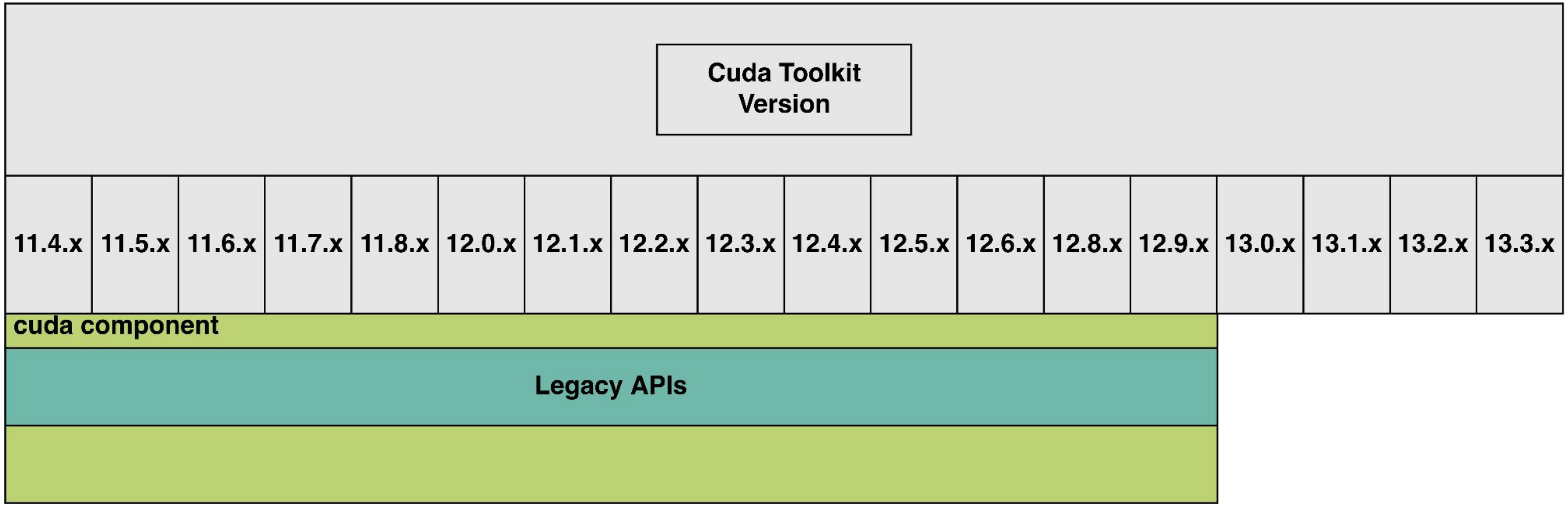

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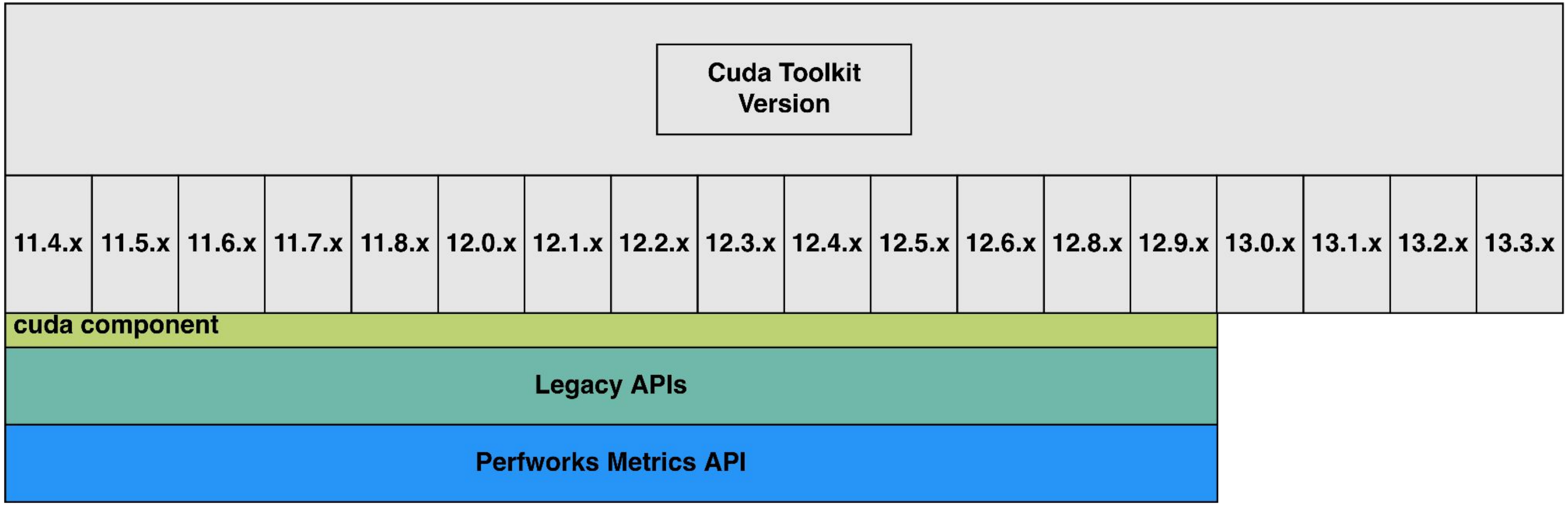

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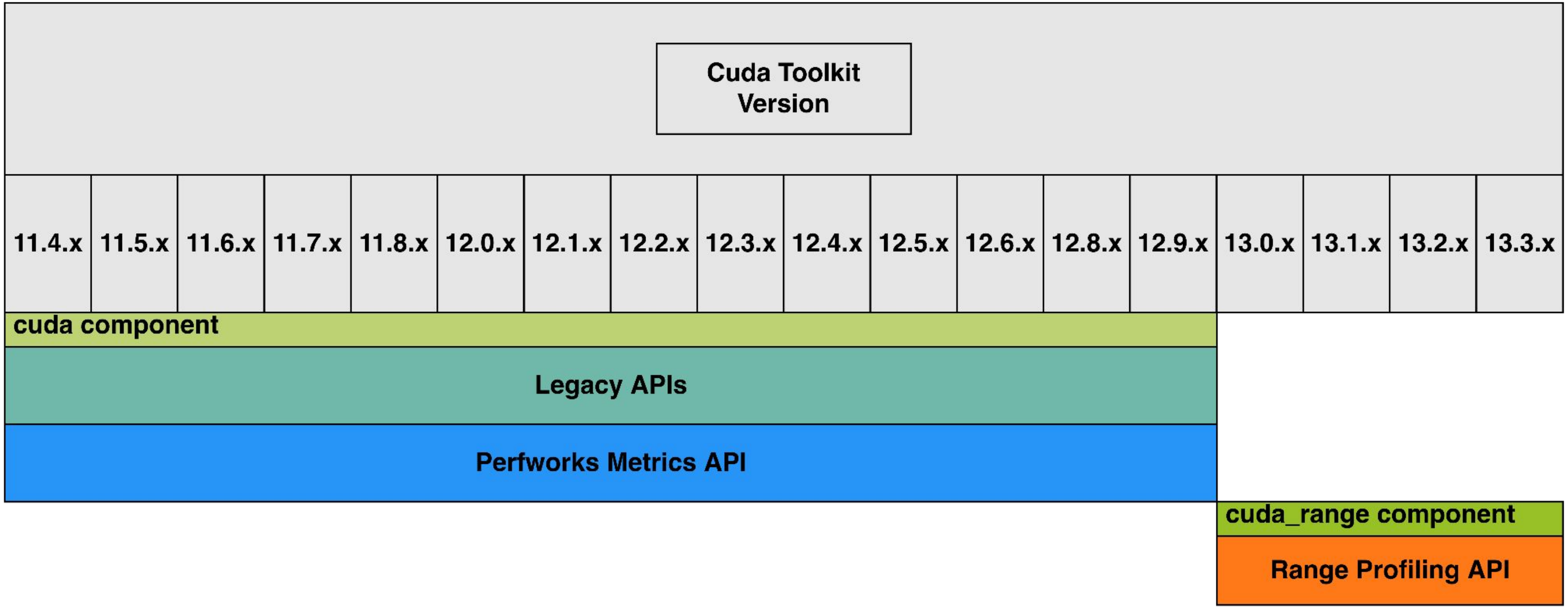

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NVIDIA Architecture and Compute Capability (CC) Support

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Architecture	CC
Kepler	3.7
Maxwell	5.0, 5.2, 5.3
Pascal	6.0, 6.1, 6.2
Volta	7.0, 7.2
Turing	7.5
Ampere	8.0, 8.6, 8.7
Lovelace	8.9
Hopper	9.0
Blackwell	10.0, 12.0

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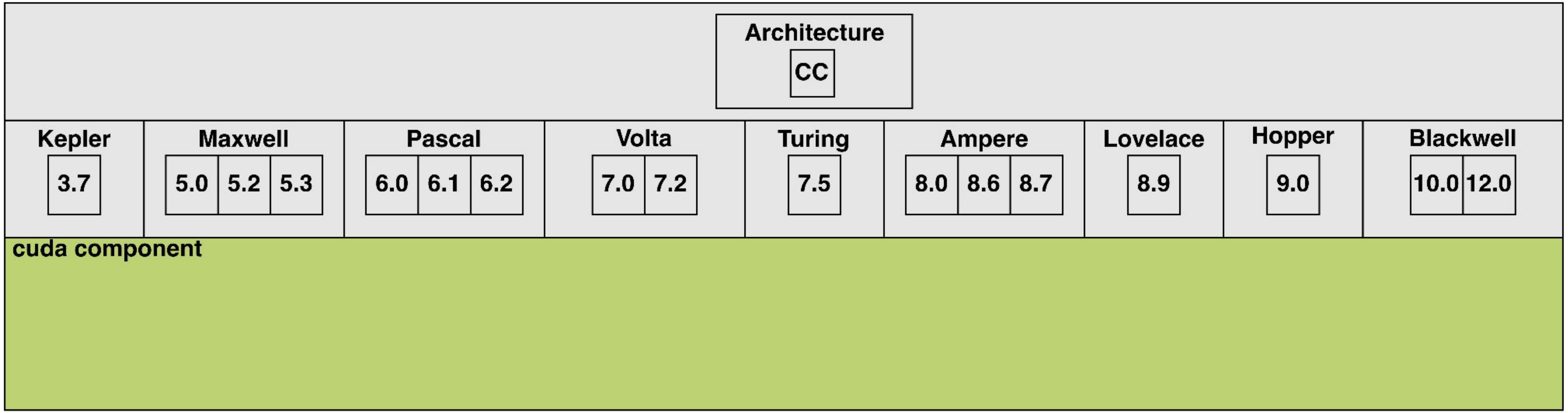

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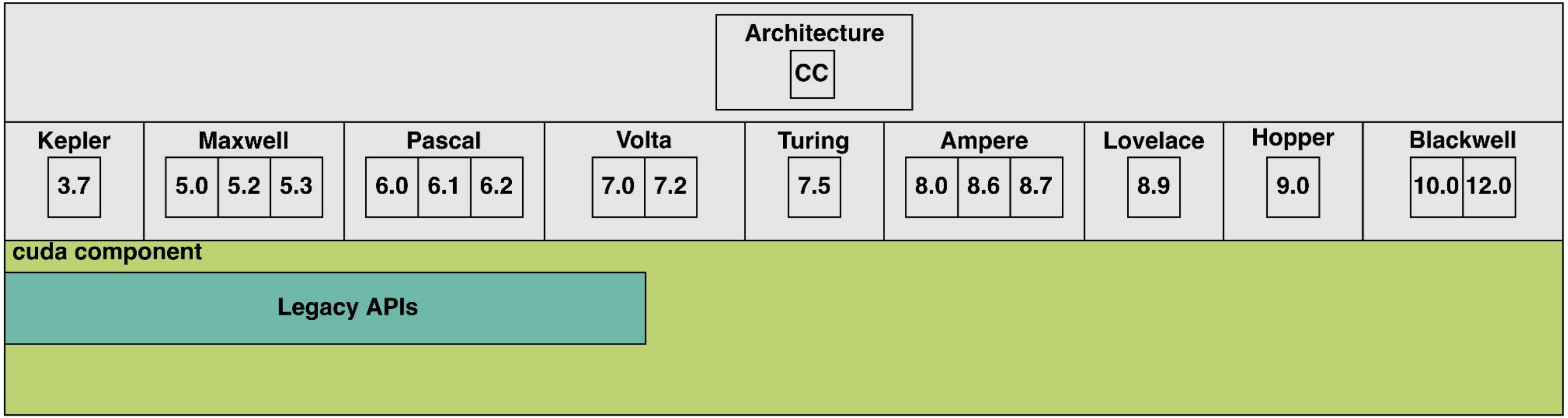

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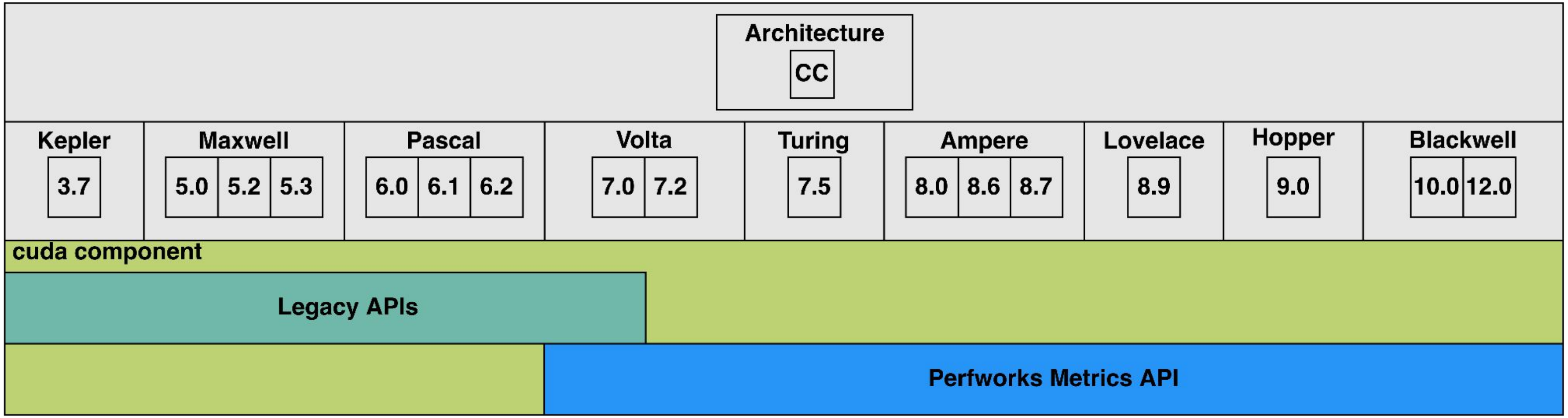

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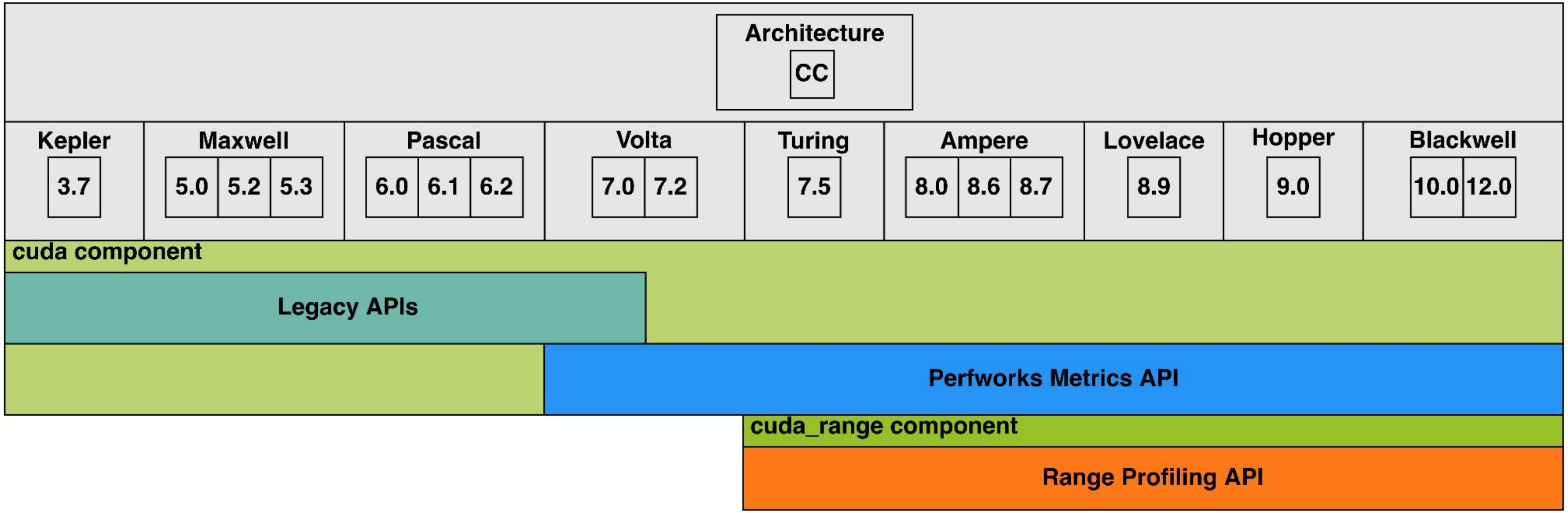

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NVIDIA GA100

Figure 4: Current Available NVIDIA GPU Presets

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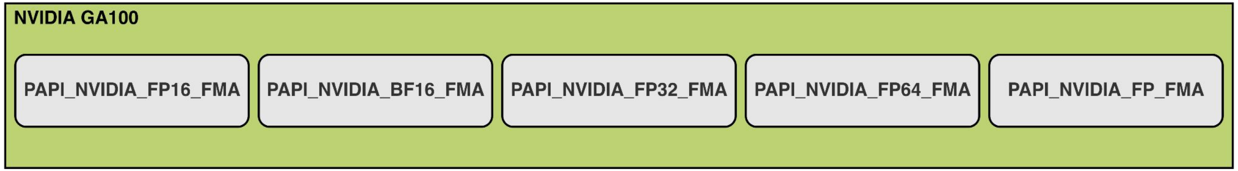

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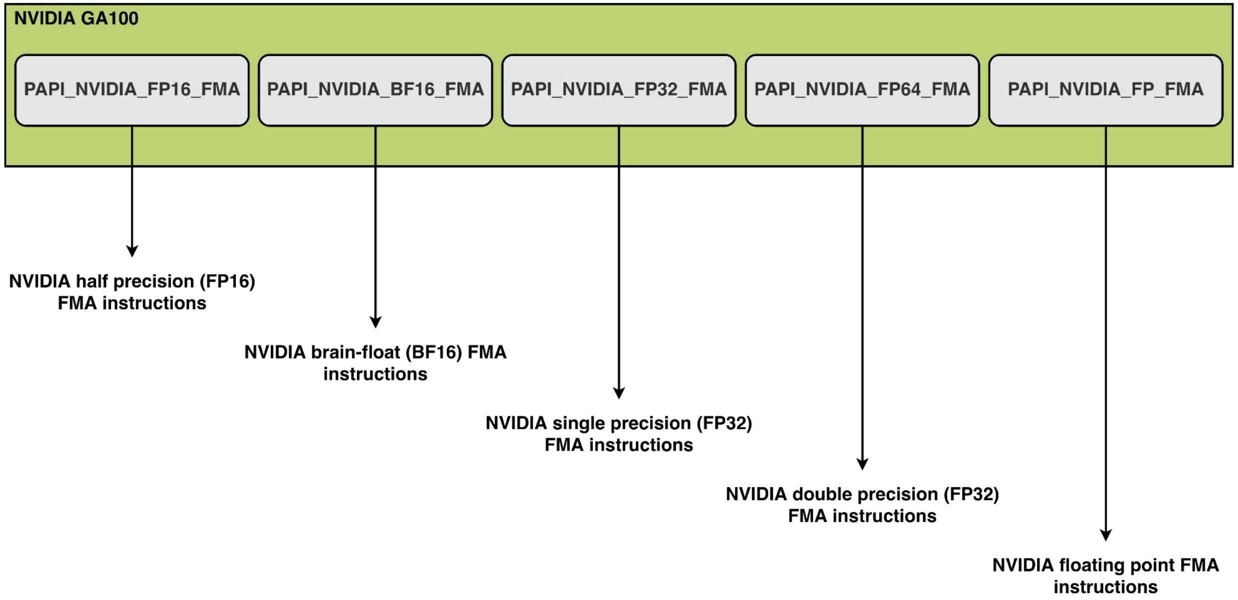

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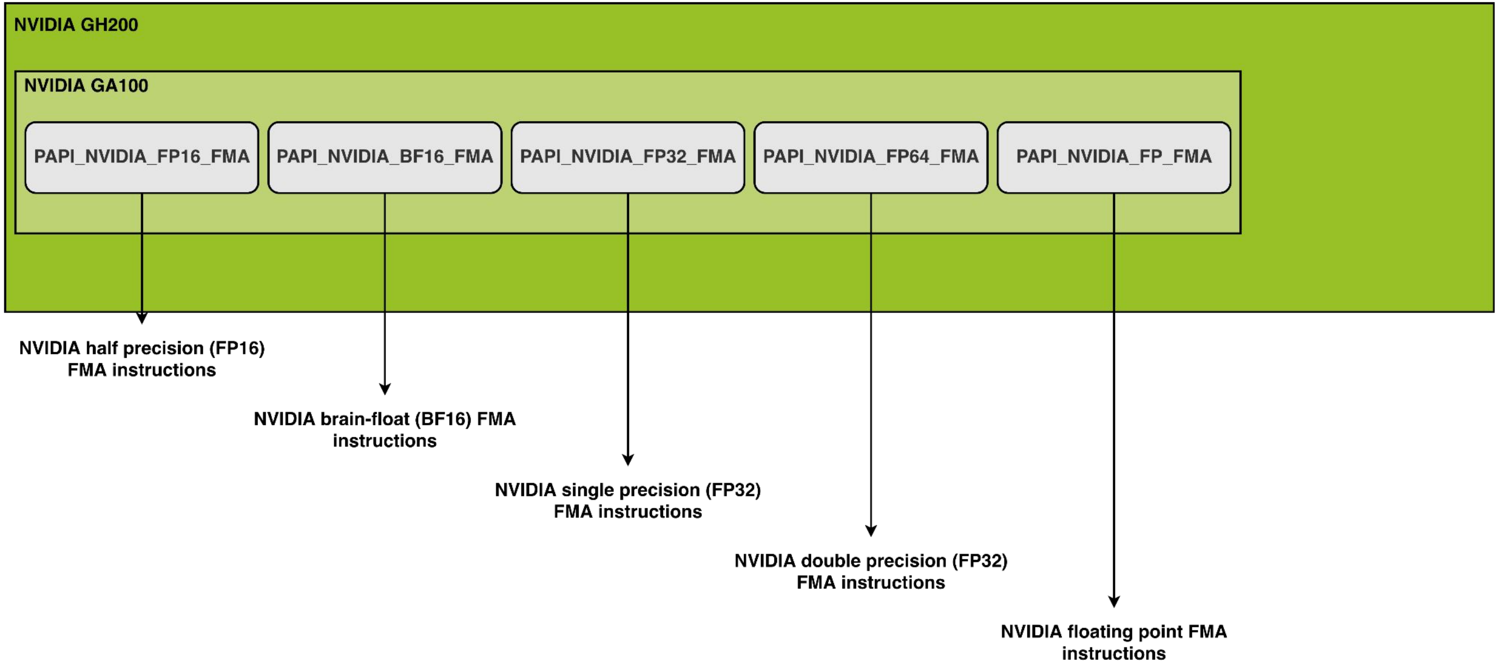

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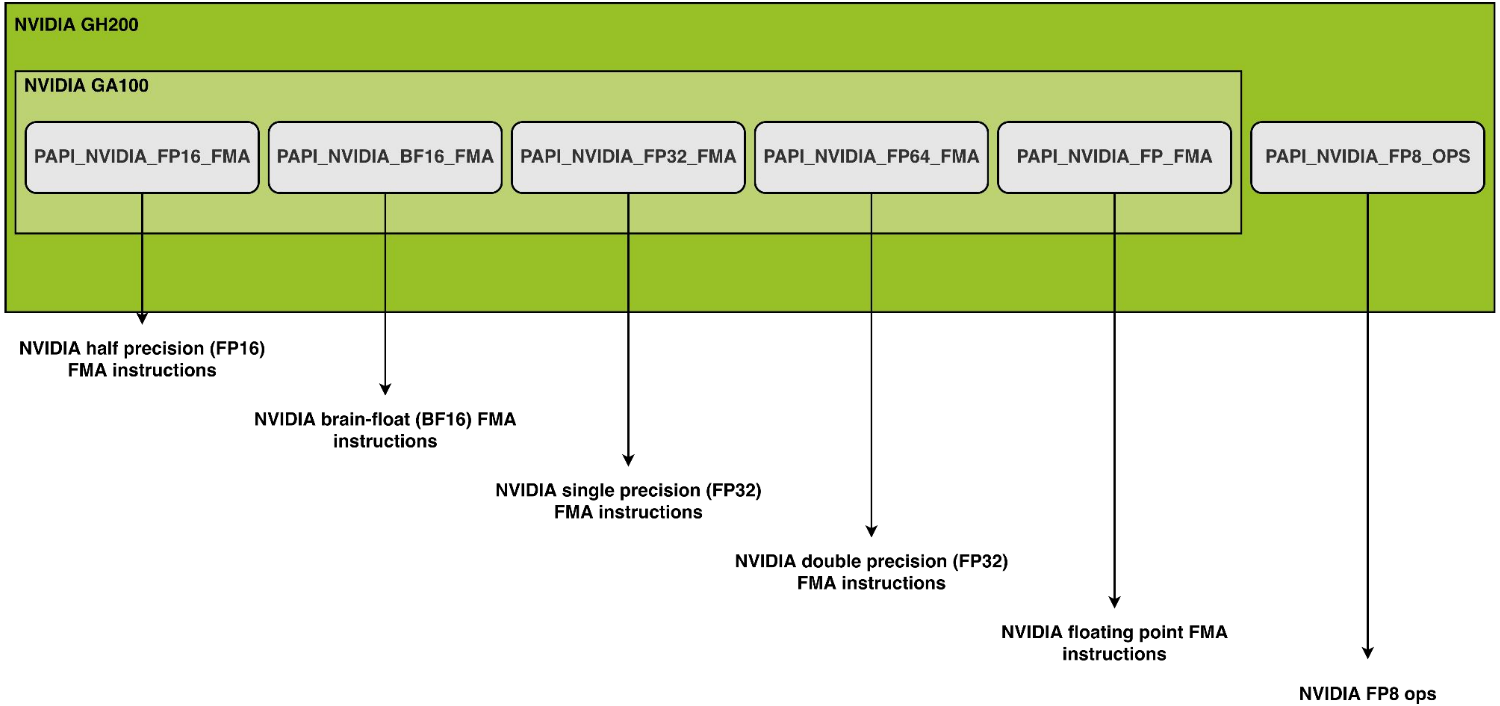

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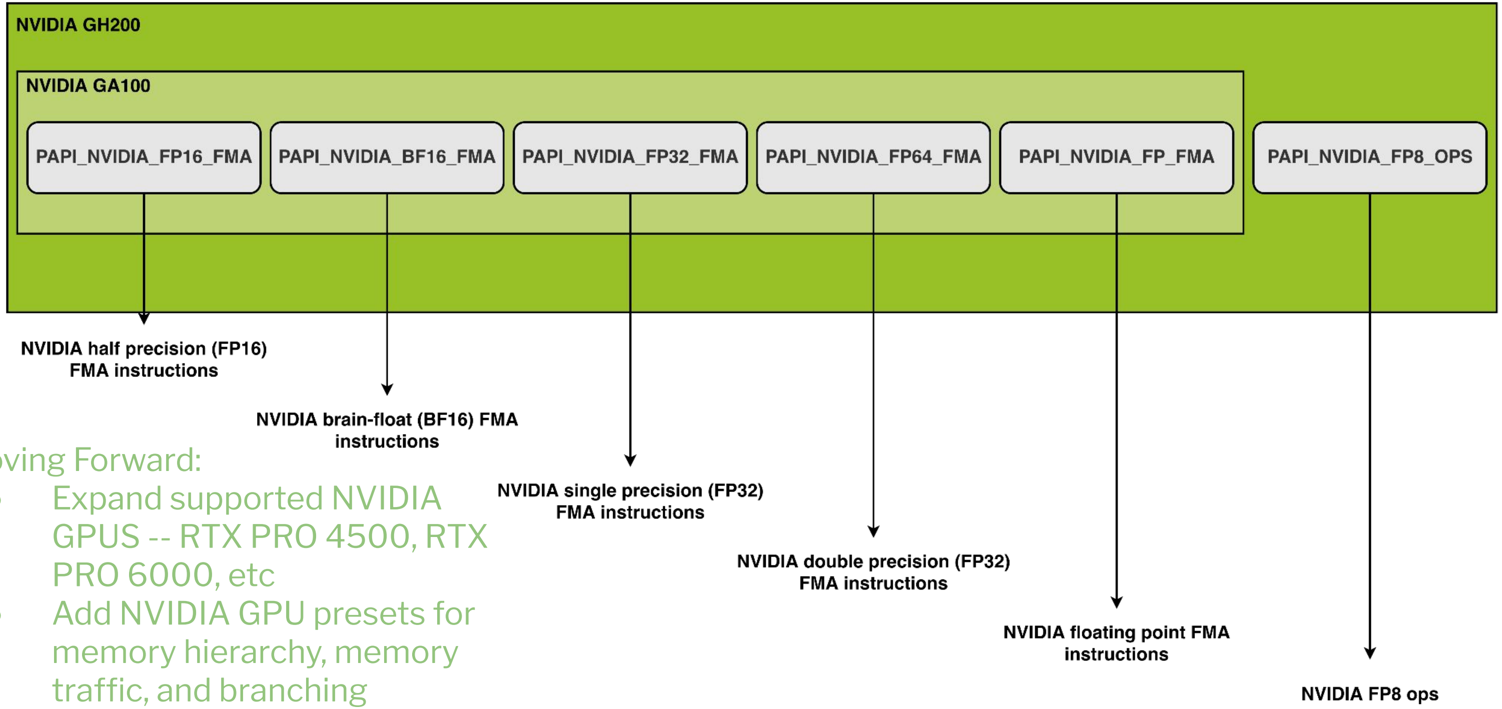

Moving Forward:

- Expand supported NVIDIA GPUS -- RTX PRO 4500, RTX PRO 6000, etc
- Add NVIDIA GPU presets for memory hierarchy, memory traffic, and branching

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Where Next for the Cuda and Cuda Range Component?

- Complete the ***cuda_range*** component for PAPI release 7.3.0
  - Exploring support for multiple pass events
- Expand support for NVIDIA GPU presets:
  - Target NVIDIA Blackwell architecture
  - Add presets beyond precision and floating point
- Performance comparison between the Perfworks Metrics API and Range Profiling API on Hopper and Blackwell architectures

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Tackling AMDs Emerging Architectures

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Overview of AMD Components

GPU
- MI250X

APU
- MI300A

Hardware Management
- Power & Frequency
- Thermal Sensors
- Interconnect Health
- ....

Figure 5: Overview of Current AMD Components

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Overview of AMD Components

GPU
- MI250X

rocm

APU
- MI300A

Hardware Management
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rocm_smi

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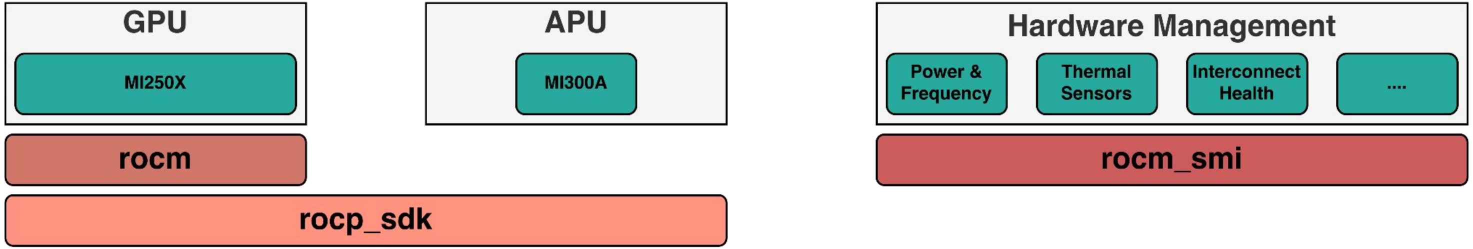
Overview of AMD Components



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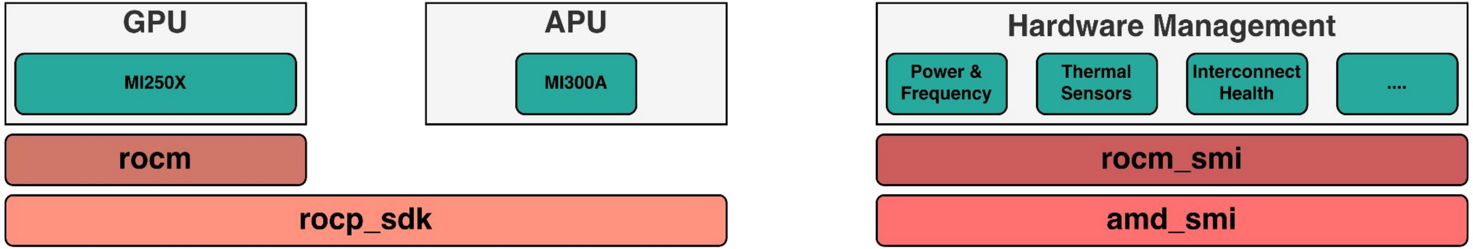

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ROCm Version Support

ROCm Version Support

ROCm Version															
5.0.x	5.1.x	5.2.x	5.3.x	5.4.x	5.5.x	5.6.x	5.7.x	6.0.x	6.1.x	6.2.x	6.3.x	6.4.x	7.0.x	7.1.x	7.2.x

Figure 6: Component Breakdown of Supported ROCm Versions

ROCm Version Support

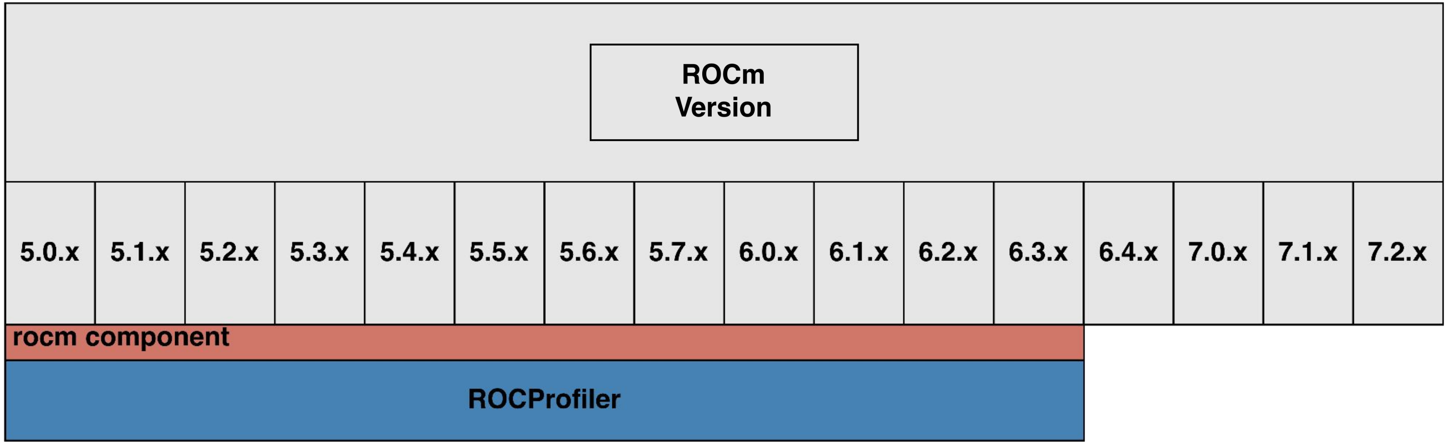
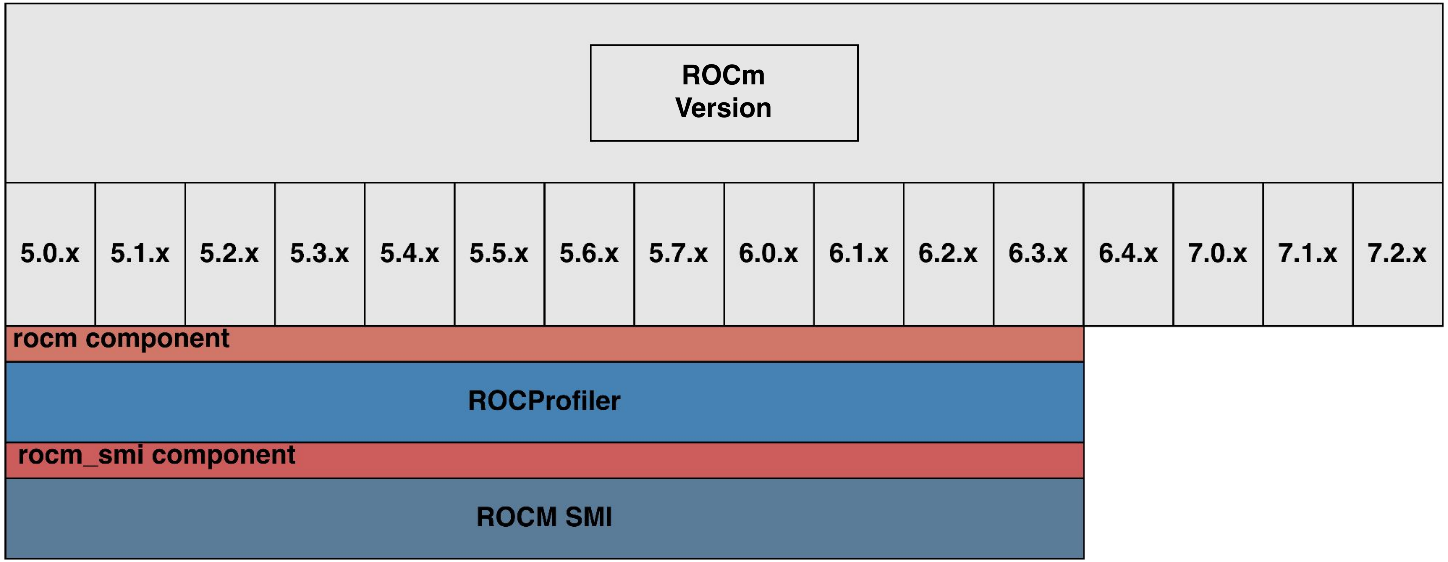


Figure 6: Component Breakdown of Supported ROCm Versions

ROCm Version Support



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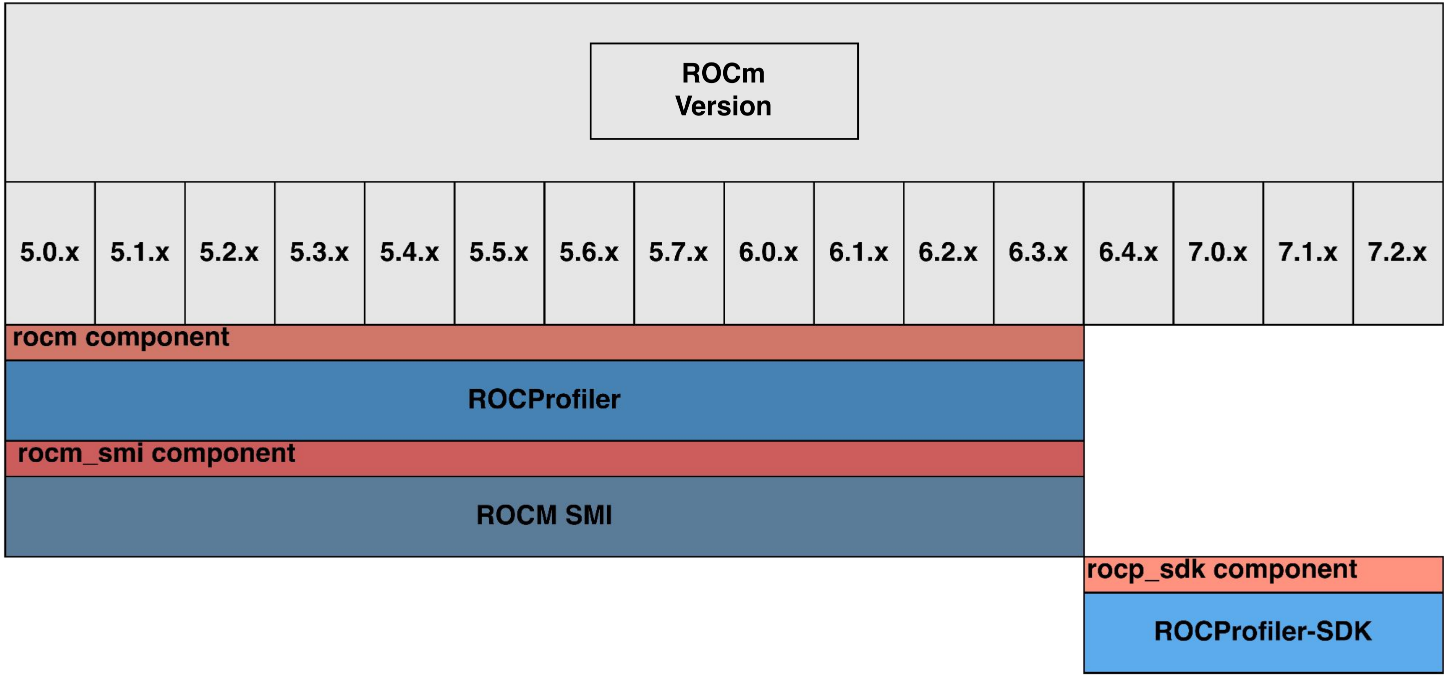

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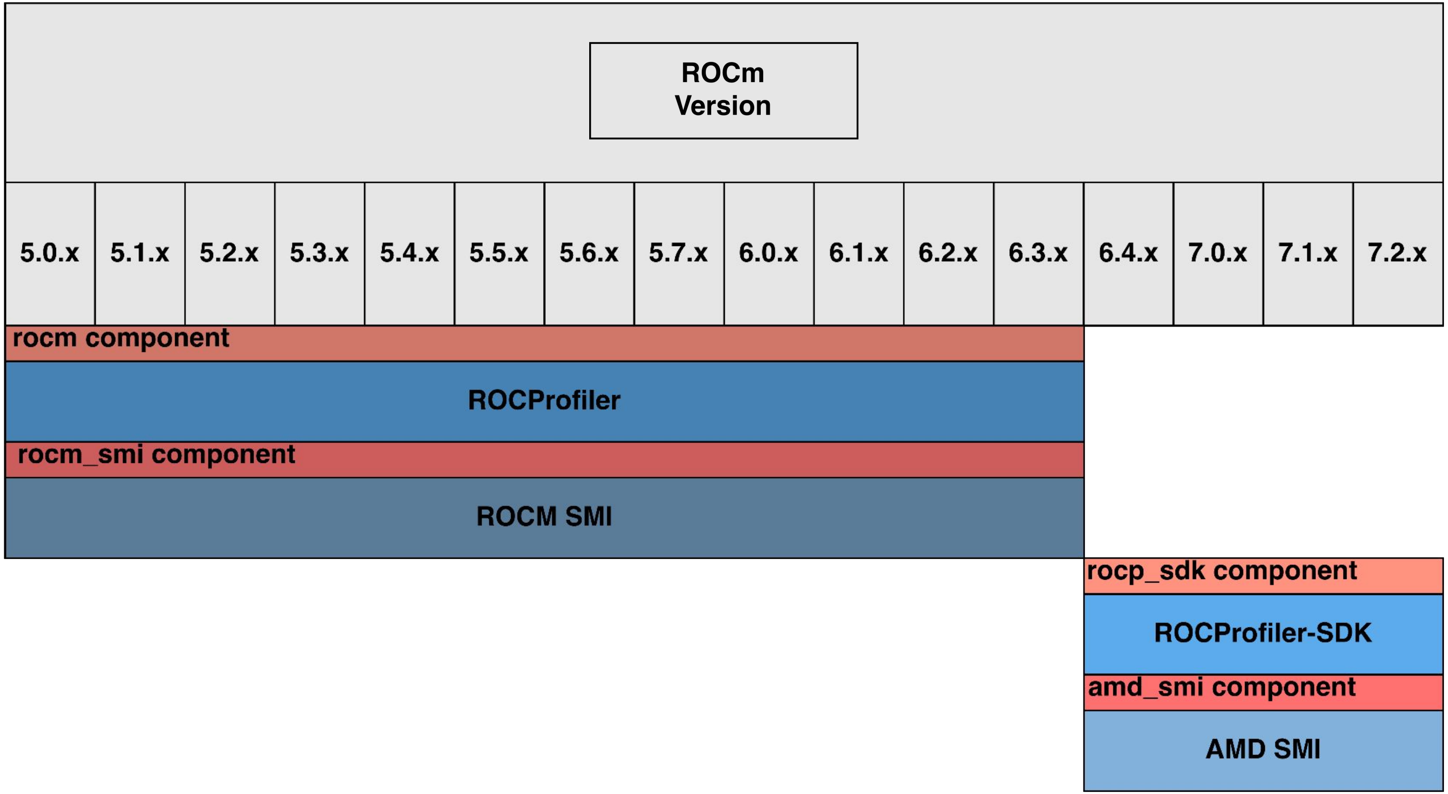

Figure 6: Component Breakdown of Supported ROCm Versions

AMD Architecture and Series Support

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AMD Architecture and Series Support

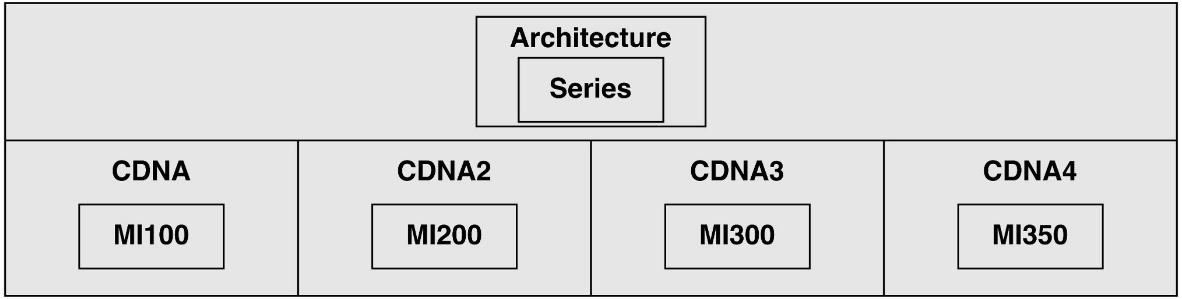
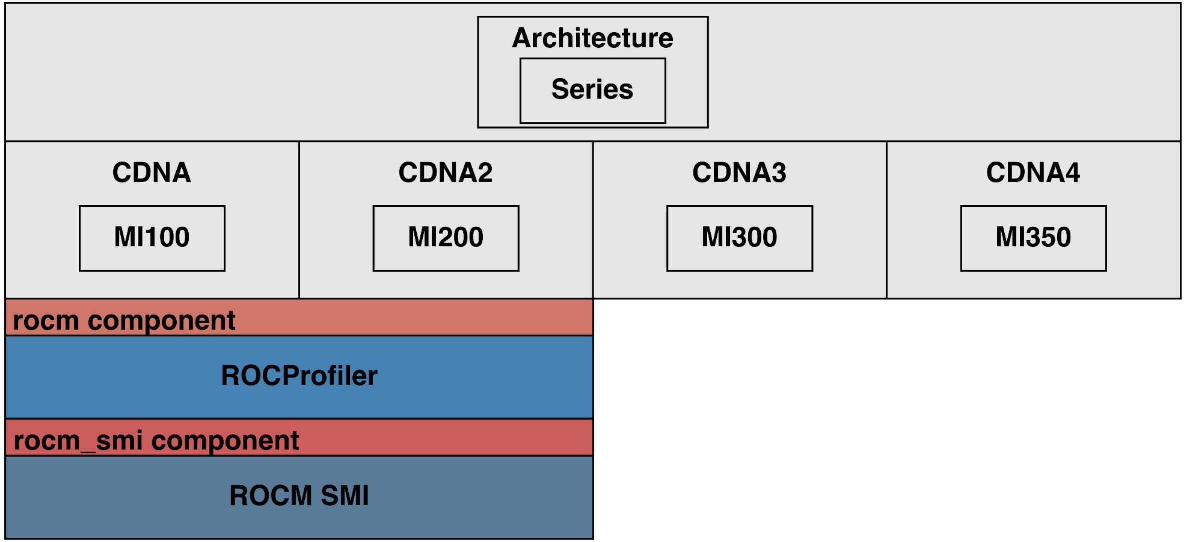


Figure 7: Component Breakdown of Supported AMD Architectures and Series

AMD Architecture and Series Support



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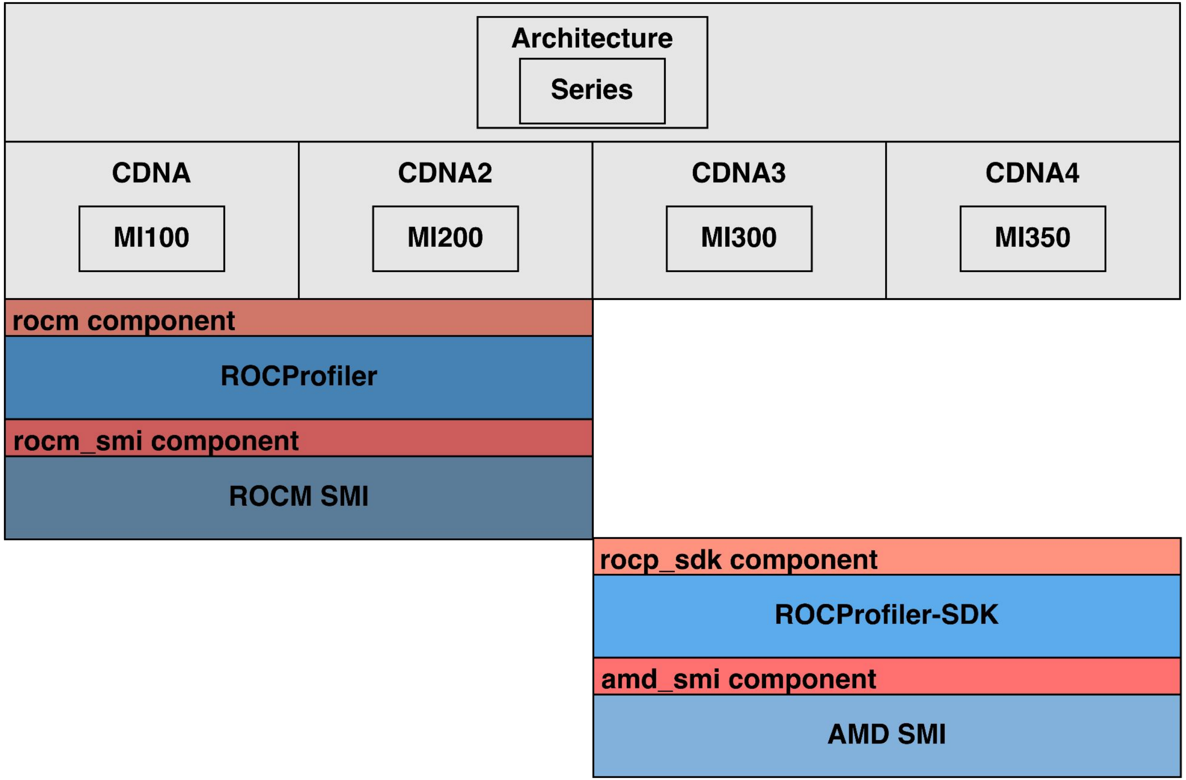

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rocm_smi vs amd_smi && rocm vs rocp_sdk…Which One?

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```
// Load the desired ROCm version.
module load rocm/7.2.4
// Configure PAPI with rocm_smi and amd_smi
./configure --prefix=$PWD/test-install --with-components=”rocm_smi amd_smi”
// Configure outputs the component to be built.
INFO: Detected ROCm version 7.2.4. Building amd_smi and skipping rocm_smi.
```

Code Block 1: Workflow for AMD Component Selection at Configure

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```

Code Block 1: Workflow for AMD Component Selection at Configure

```
// Run papi_component avail.
./papi_component_avail
// Output of papi_component_avail.
Name:   perf_event              Linux perf_event CPU counters
Name:   perf_event_uncore       Linux perf_event CPU uncore and northbridge
Name:   amd_smi                 AMD GPU System Management Interface via AMD SMI
library
```

Code Block 2: Output of ***./papi_component_avail***

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AMD GPU Presets

AMD GPU Presets

- AMD GPU presets are tentatively planned for PAPI 7.3.0

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AMD MI250X

Figure 8: Tentative AMD GPU Presets

AMD GPU Presets

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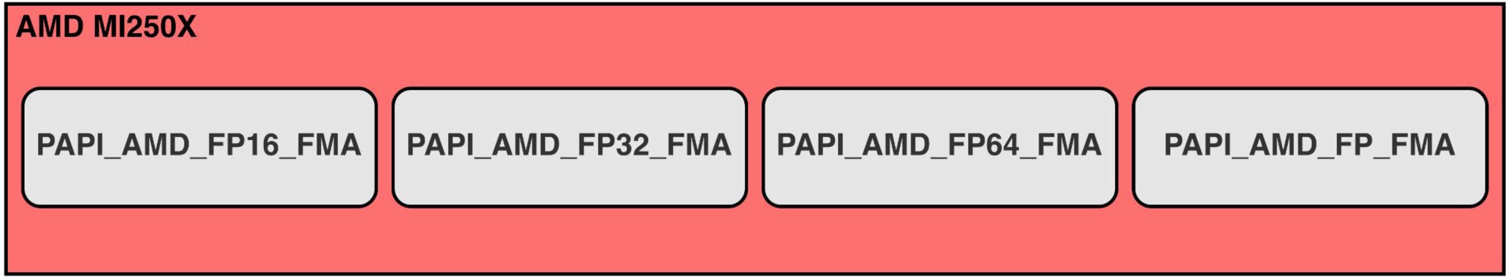

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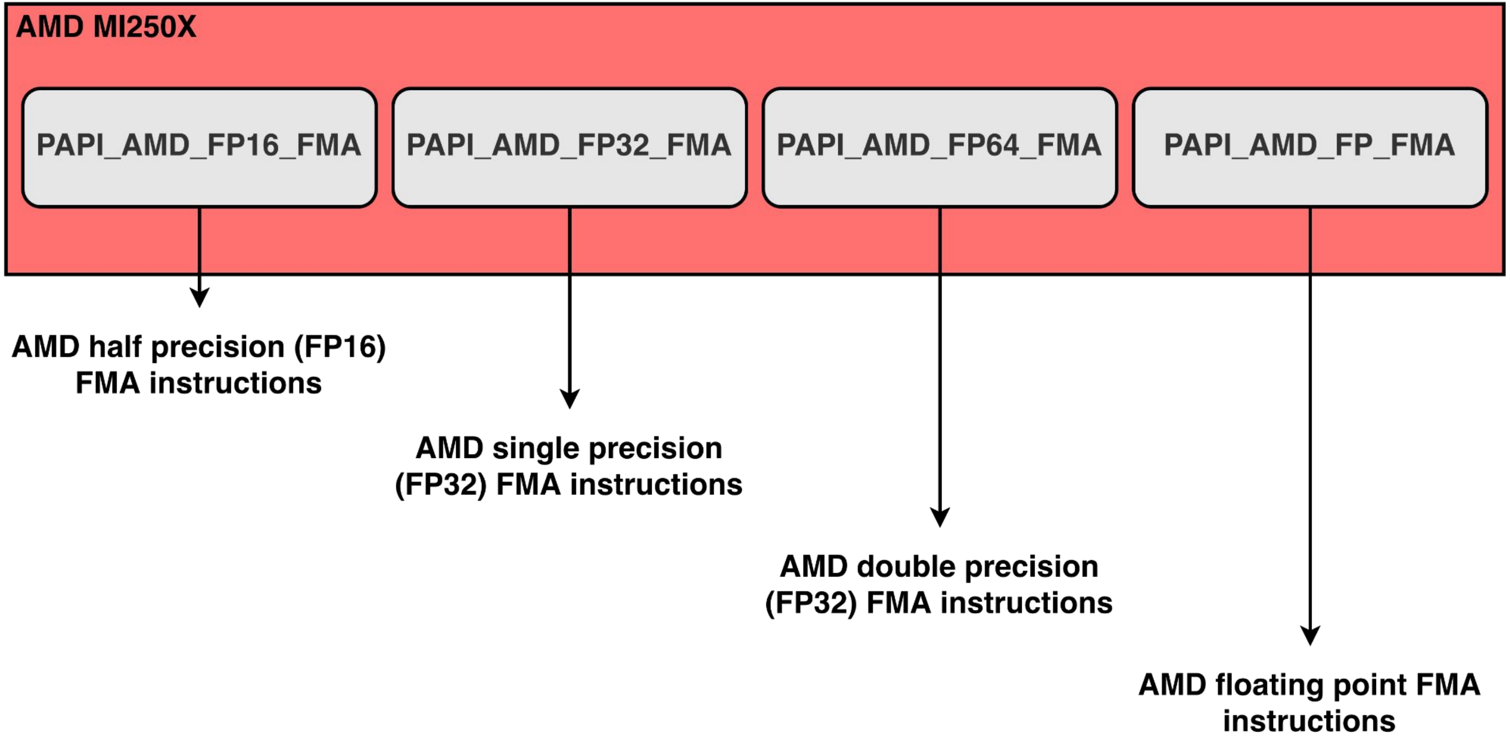

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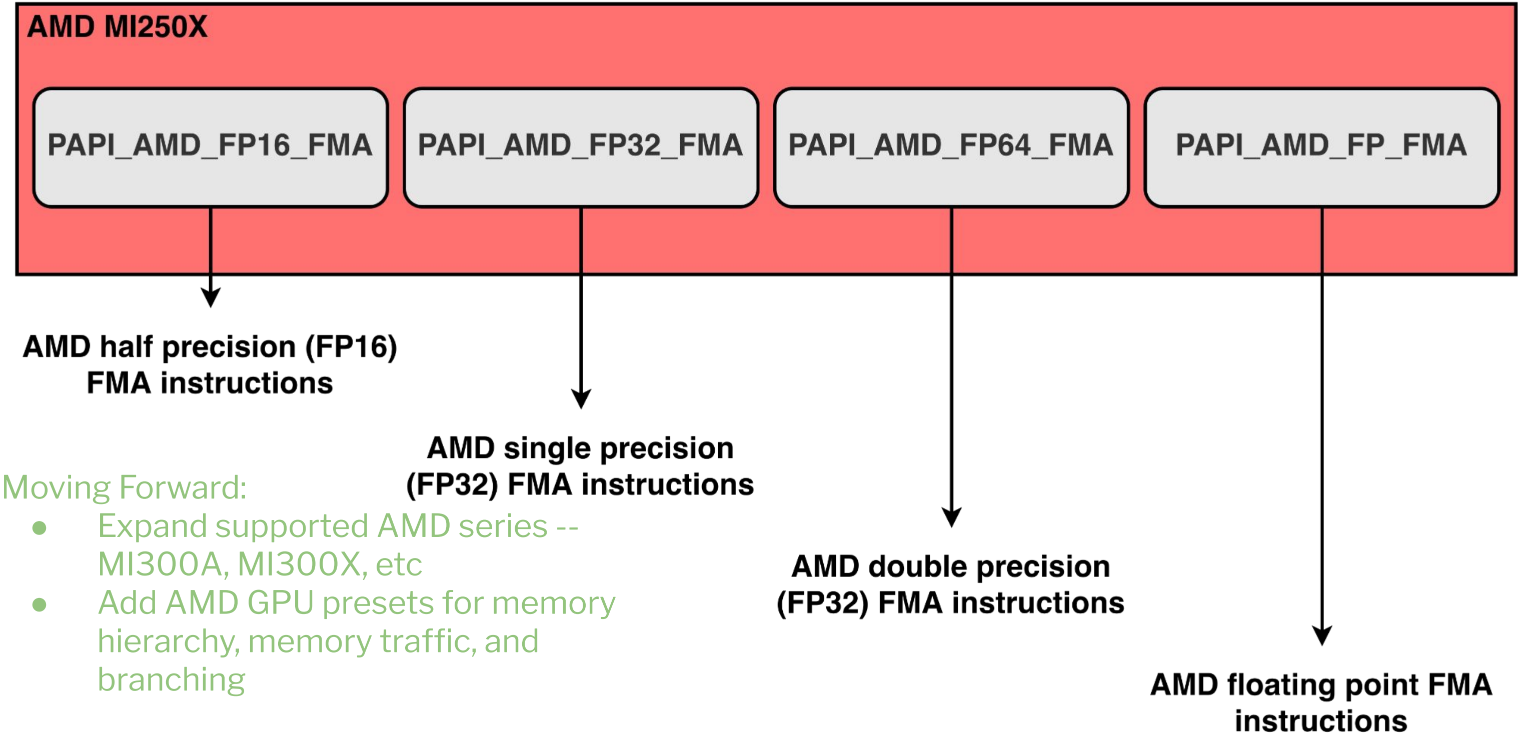

Moving Forward:

- Expand supported AMD series -- MI300A, MI300X, etc
- Add AMD GPU presets for memory hierarchy, memory traffic, and branching

Figure 8: Tentative AMD GPU Presets

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PAPI Continuous Integration

Enabling the Latest Architecture and Version Support

- PAPI aims to support legacy and modern software/hardware releases for AMD, Intel, and NVIDIA; as a result, manual testing is cumbersome

Software Releases	Hardware Releases
+ Compilation errors (e.g. API naming schemes changed or structures introducing new member variables) + Dropped architectures (e.g. Cuda Toolkit 13.0.0 dropping offline compilation for per-Turing architectures)	+ Outdated and unsupported APIs (e.g. CUPTI Legacy APIs capped at CC 7.0) + Required driver versions

Figure 9: Possible Issues Encountered on Software/Hardware Releases

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- Oregon Advanced Computing Institute for Science and Technology (OACISS) allows PAPI to run continuous integration (CI) on legacy and modern software/hardware releases to detect early:
  - Bugs
  - Incompatibility between software and hardware

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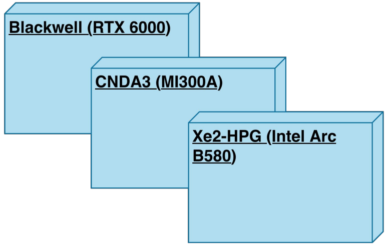

Figure 10: Subset of GPU and CPUs Available Through OACISS

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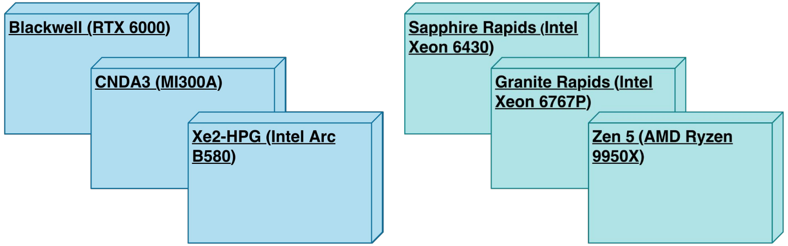

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